Protein metabolism: Since diabetic patients cannot meet the energy requirements from the oxidation of glucose and fats, there is breakdown of tissue proteins. There is increased protein catabolism leading to negative nitrogen metabolism.

9.6. Diagnosis of Diabetes

Several laboratory tests are available to both health care providers and people in order to confirm a diagnosis of Diabetes.

Fasting Plasma Glucose (FPG) Test: The Fasting Plasma Glucose Test involves asking the person to abstain from eating or drinking anything for eight hours prior to having a blood sample drawn. If the person's glucose level while they are fasting is greater than or equal to 126 mg/dl they are most likely to have diabetes. If the person has a fasting glucose level that is more than 100, but less than 126 mg/dl - they are considered to have Pre-Diabetes. They do not yet have Diabetes, but they are at greater risk of developing it in the near future.

Oral Glucose Tolerance Test: An Oral Glucose Tolerance Test involves first drawing the person's fasting blood sugar level, and then drawing another blood sugar level two hours after they have consumed a drink containing seventy-five grams of sugar. If the person's blood sugar level after they have consumed the sugar drink is greater than or equal to 200 mg/dl they have Diabetes. If their blood sugar level is between 140 and 199 mg/dl, they are considered to have Pre-Diabetes.

Finger Stick Blood Glucose Test or Random Plasma Glucose (RPG) Test: This test can be performed in community-based screening programs, or anywhere, and provides rapid results. The test is not as accurate as blood testing in a laboratory; however, it provides rapid results, and it is very easy to perform. The results are accurate within ten-percent of laboratory values. Testing involves a minor finger-stick of the person's finger in order to obtain a tiny blood sample, which is then placed onto a strip. The strip is placed into a small machine that interprets the person's blood sugar level. At very high or low blood sugar levels, the finger-stick test may prove inaccurate, and is considered a preliminary screening. The majority of persons with diabetes use the finger-stick test to monitor their blood sugar levels.

The following image shows small diabetes blood sugar meter - reading 4.8 - a finger pricker, syringe, and blood glucose test strips.



Monitoring Urine Glucose and Ketone levels

Glycosuria is usually measured by

- Clinitest (modification of Benedict's test)
- Clinistix, Testape, Diastix (Paper strip method)

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